

cycles and large sizes of gymnosperms have made traditional, breeding-based analyses of these plants challenging. DNA-based technology that can bypass these limitations has been particularly useful in forest trees, enabling genomic mapping, gene sequencing, genomic selection and genetic engineering. Whole-genome sequences are particularly powerful, because they provide a platform for a spectrum of new technologies, such as studies of an organism's full transcriptome — its complement of transcribed RNA molecules. Nystedt and colleagues' draft sequence is the first gymnosperm sequence to be published (Fig. 1), and a loblolly pine genome sequence (around 22 Gb) is expected to follow soon⁵. These conifer genomes are the largest plant genomes sequenced so far, much larger than the 17-kb genome of wheat⁶ (an angiosperm).

The spruce genome will not only accelerate the investigation of gymnosperm biology, it will also provide broader genetic and evolutionary insight. For example, researchers of the ENCODE project⁷ recently argued that 70% of the 3.2-Gb human genome is functional in some way. But large plant and animal genomes pose a challenge to this proposal. Consider, for instance, the spruce genome (around 20 Gb) and that of thale cress, the model plant *Arabidopsis thaliana* (0.135 Gb). If a similarly high proportion of these genomes were functional, what properties of gymnosperms could necessitate such a massively larger number of functional genetic elements? Repression of only two genes enables *A. thaliana* to extend its growth cycle and produce a substantial amount of wood⁸, suggesting that the large genomes of gymnosperms are not attributable to a perennial growth habit or wood formation. Early comparisons of transcribed genes indicated⁹ that the vast majority of genes in gymnosperms and angiosperms have homologues in the same gene families. Nystedt and colleagues estimate that the spruce genome contains around 28,354 genes, which is very close to *A. thaliana*'s 27,407. The angiosperms maize (corn), rice and poplar have estimated gene numbers of around 40,000. But these coding regions comprise a fraction of the sequence of these large genomes, and what the functions of the remaining sequences may be is still obscure.

Another curious aspect of gymnosperm genomes is the evolutionary conservation, in many species, of a haploid (single copy) chromosome number of 12, despite their genome sizes ranging from 9.7 to 37 Gb¹. The genomes will also provide insight into mechanisms of ancient and recent evolutionary adaptation in plants. Gymnosperms are thought to have originated from progymnosperms 360 million years ago, but much about the origin of angiosperms remains a mystery. Although the fossil record supports a gymnosperm origin for angiosperms, estimates of the time

of angiosperm origins vary by more than 100 million years. There is also controversy regarding relationships within the gymnosperms, particularly the relationship of the conifers to the gnetophytes, which include the bizarre desert plant *Welwitschia*.

The genome sequences of these trees will not only help us to understand the past, but may also increase our understanding of present-day northern-latitude forests. Gymnosperms became the dominant forest plants in the late Palaeozoic and the Mesozoic periods, around 300 million to 70 million years ago. But during the most recent ice age, much of the northern latitudes were covered by ice. When the glaciers last retreated, only about 10,000 years ago, the conifers were the major pioneer species that dominated that land. Understanding how the gymnosperms established new ecosystems as the glaciers shrank is becoming more important as we anticipate the

effects of global climate change on the world's forests. ■

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PLANETARY SCIENCE

Evolutionary dichotomy for rocky planets

A simple model shows that a rocky planet close to its star may solidify so slowly that its water is lost to space and the planet becomes desiccated, whereas a planet farther out may solidify quickly and retain its water. SEE LETTER P.607

LINDA T. ELKINS-TANTON

Earth and Venus were probably built from similar rocky materials, having been formed by similar mass-accretion processes. The giant impacts that characterize these processes are thought to melt the

growing planets to some depth, producing one or more magma-ocean stages during which the silicate portion of the planet is melted before solidifying. Thus, there has been no reason to suspect that Venus and Earth differed through the first tens and probably hundreds of millions of years of the Solar System, and Venus

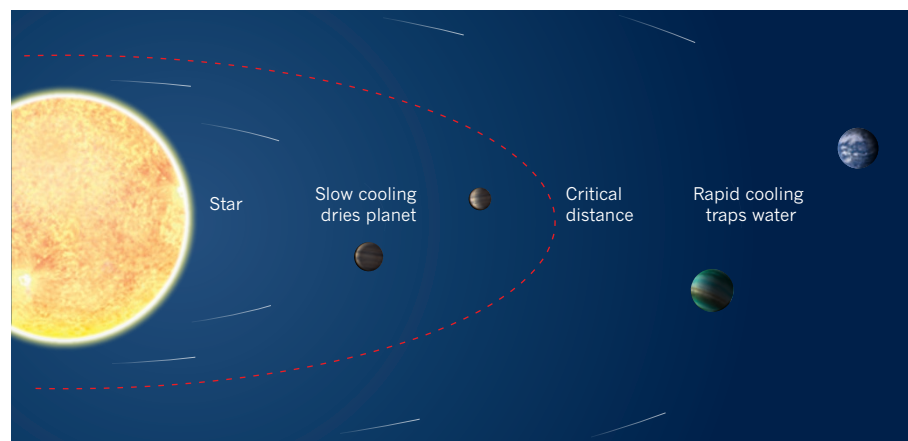


Figure 1 | Hamano and colleagues' evolutionary model for terrestrial planets¹. Rocky planets that form and solidify close to their host stars may receive enough heat from the star to slow cooling, allowing time for water to escape into space and causing the planet to dry out. Rocky planets that form farther from their stars, beyond a critical distance, would cool quickly, trapping water in silicate minerals in their interiors and as liquid on their surfaces.

is commonly thought to have lost its water through some later divergence from Earth-like evolution. On page 607 of this issue, Hamano *et al.*¹ present a simple model that might explain why rocky planets that have similar compositions but orbit at different distances from their host stars can end their magma-ocean stages with either an Earth-like wetness or a Venus-like dryness. This model does not require any later divergence to explain the differences between the planets.

Almost 30 years ago, researchers showed how a dense steam atmosphere can be generated on a young, hot planet by the solidification of an impact-generated magma ocean². After the upper troposphere (the lowest portion of the atmosphere) of the young planet has become saturated with steam, that atmospheric layer imposes a strict upper limit on outgoing radiation from the magma ocean — about 300 watts per square metre. Therefore, as soon as the magma ocean produces a steam-saturated troposphere, the cooling rate of the planet is controlled by this one simple limit.

Previously, several groups had calculated that a magma ocean should solidify in just millions of years^{3–5}. These calculations assumed that the planet had lower incoming heat flux from the star than outgoing heat flux from the magma ocean. The crucial feature of Hamano and colleagues' model is that some planets are close enough to their star for the incoming heat flux to be higher than the 300 W m^{-2} outgoing radiation limit, and thus the planet would be prevented from cooling at all until water was lost from the steam-saturated atmosphere.

For planets close to their star, solidification and cooling together may take orders of magnitude longer than for more distant planets, perhaps as long as hundreds of millions of years. After solidification, cooling proceeds only as water is stripped from the hot, inflated atmosphere by hydrodynamic escape. The longer that solidification and cooling take, the more water is lost to space, and the drier the planet becomes. Thus, the distance of a terrestrial planet from its host star might produce an evolutionary dichotomy (Fig. 1).

The authors further suggest that Earth solidified far enough from the Sun to have a net loss of planetary heat from the beginning, allowing it to solidify quickly. Earth's initial water inventory influenced the volume of only its initial oceans. Venus, however, may have had net heat flux into the planet, and its current dryness might be related to this early slow solidification and attendant atmospheric water loss, before cooling allowed the water in the steam atmosphere to cool and condense into liquid oceans.

Recent work on geochemical tracers has indicated that Earth's mantle solidified and differentiated from a magma ocean more than 4.45 billion years ago⁶, probably around 4.52 billion years ago⁷, which agrees with rapid

solidification. For Venus, however, there are insufficient geochemical data to perform this test. Measurements of deuterium and hydrogen in the Venusian atmosphere indicate that the planet has lost a substantial amount of water over time^{8,9}, but whether that loss occurred at the time of solidification or more recently is a matter of argument.

The authors' model underscores the importance of the earliest accretion and solidification steps in determining the future evolution of the rocky planets. However, several crucial caveats need to be considered in applying this model. First, in extrapolating back in time, the faint young star's radiation level needs to be considered. Second, initial atmospheres might not all be water-rich; the rocky building blocks for some planets might have produced atmospheres rich in methane and hydrogen, instead of steam¹⁰. In the absence of a steam atmosphere, there would be no outgoing radiation limit to slow solidification and cooling. Third, forming an initial atmosphere above a magma ocean is not a simple process. The removal of volatile gases from magma might require a significant degree of supersaturation and might not occur until late in solidification. If this is so, then solidification would proceed to a high degree before a steam atmosphere formed and occluded heat flux.

Although proximity to a star affects planetary water content, this is not the only parameter that dictates the habitability of a rocky planet — the planet's composition also has a strong influence on all aspects of habitability, such as bulk atmospheric composition, susceptibility to plate tectonics and formation of a shielding magnetic field. A challenge for the coming decades will be to make measurements of exoplanets that allow the testing of models for habitability, and these tests need to include composition. How do atmospheric species other than water affect the solidification rates of magma oceans? What atmospheric compositions would be expected in the wake of a slow solidification with substantial water loss? The habitability of Earth and the inhospitability of Venus may be the inevitable result of our planetary sibling order next to the Sun rather than later evolutionary bifurcations. If so, similar patterns of habitability are likely to be found in exoplanets. ■

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digital systems, the design of analog circuitry requires greater expertise in feedback-system design, circuit theory and signal processing. Daniel and colleagues' work is founded on the close analogy they saw between the exponential thermodynamic electron flow that occurs in transistors and the exponential thermodynamic rates seen in chemical reactions⁵. This similarity led them to realize that analog electronic circuits that operate in the logarithmic domain might be effective mimics of analog biological circuits that operate in the logarithmic domain.

To test this idea, Daniel *et al.* constructed a gene-transcription unit in which the concentration of a transcription factor (a protein that regulates gene transcription) could be logarithmically transformed over a wide range of concentrations, thus enabling fine control of gene expression.

Moving an input signal into the logarithmic domain has many benefits, especially that basic calculations such as division and multiplication can be more easily executed in log space. The core of the authors' logarithmic unit is a positive-feedback loop operating in an analog mode. A shunt is included that removes excess transcription factor so as to increase the range of operation and efficiently implement positive feedback. With this basic logarithmic unit in place, the researchers could then focus on adding higher-order functions. For example, by attaching a repressor module, they created a circuit whose readout was the negative of the log of the input. Another circuit simply summed the activity of two parallel logarithmic circuits to mimic a multiplication operation.

Furthermore, using just two transcription factors, the authors were able to create a negative-feedback analog circuit that scaled the logarithmic function. Especially interesting was their construction of a circuit that computed the ratio of two inputs over four orders of magnitude of input concentration. Computing a ratio might have many applications, such as normalizing or comparing values, or even providing an *in vivo* pH meter.

The ability to process graded information by means of synthetic biological circuits will be of interest to many researchers. For example, most inputs from the environment are graded, and a synthetic circuit with a fluorescent protein readout might be able to represent the rate of change of an environmental input rather than its absolute level. This could be accomplished by having an analog differentiator circuit. Other analog designs might be used to measure the weighted sum of environmental inputs, such that when a particular combination of inputs is reached it triggers a change in the state of the cell. Designing such analog circuitry may also further our understanding of natural systems. The function of many regulatory systems in cells remains obscure, and studies of synthetic biology in the analog domain should lead to

SYNTHETIC BIOLOGY

It's an analog world

The first synthetic genetic circuits to use analog computation have been developed. These circuits involve fewer components and resources, and can execute more complex operations, than their digital counterparts. SEE LETTER P.619

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The engineering of biological networks, a discipline called synthetic biology, has seen remarkable progress since its inception in 2000. The fact that we can now design simple but functional circuits *in vivo* is probably one of the most important aspects of this progress — it suggests that our biophysical understanding of the cellular milieu is largely correct, even if many details remain to be resolved. The significance of this should not be underestimated, because it means that predictive engineering of new cellular networks and systems is possible. Until now, approaches to synthetic biology have been greatly influenced by the digital computing technology that we use every day. But biological cells do not process information in a solely digital fashion; rather, they carry out many operations in an analog manner. In this issue, Daniel *et al.*¹ (page 619) break the mould and venture into a completely new area of synthetic biology, presenting synthetic circuitry that can perform analog calculations*.

Synthetic biology has already led to several notable firsts — from the construction of switches, oscillators and feed-forward networks to the creation of gene regulatory circuits that can carry out Boolean logic. In many ways, digital computing has coloured our approach to synthetic biology. It is a truly remarkable achievement that so much can be done with just the 'ones' and 'zeros' that form the basis of digital processing. After more than 60 years of digital computing, it is no surprise that many modes of thinking have become highly influenced by the digital paradigm. It is often said that a biological cell is just like a digital computer, and that reprogramming a cell is like writing software. But the digital analogy is a misleading one, because many processes that occur in a biological cell have no counterpart in a digital computer. If reprogramming a cell was just a case of writing code, then re-engineering cells would be simple — but it is not.

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The reality is that cells use a hybrid approach to information processing. In some cases they use digital yes-or-no decisions, but in many cases cellular signals are analog, with levels of gradation. More exotic, little understood signal-processing techniques involving noise and other forms of signal probably also contribute. And on top of that, a complex chemistry exists that continuously reassembles the cell in real time.

Nevertheless, constructing digital devices is both a useful and an interesting engineering challenge for synthetic biologists, and many advances have been made in mimicking digital

“The ability to process graded information by means of synthetic biological circuits will be of interest to many researchers.”

systems either *in vivo* or *in vitro* using DNA as components. One remarkable study was the construction of an *in vitro* 4-bit square-root calculator consisting of 130 DNA strands². Another was the construction of an *in vivo* system to detect four

different inputs, comprised of four sensors and three logic gates³.

Although these are significant achievements, such circuits require large numbers of components to perform even the simplest computation. A 4-bit binary adder, for example, might require 30 or more proteins to operate, and at the same time would place a substantial metabolic burden on a cell. More challenging and perhaps more interesting is to attempt to mimic analog computation in a cell, as Daniel *et al.* have done.

One key advantage of analog over digital is that far fewer devices are needed to carry out a given computation at the moderate precision needed in cells⁴, and fewer devices mean lower resource requirements. In addition, the richness of signal processing that can be carried out in analog systems is far greater than what can be accomplished in digital systems using the same number of components. But analog information processing brings with it a new set of challenges. Compared with that of